

STATS 201A Lecture 2

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29 September 2026

$$\varepsilon_{ij} \stackrel{iid}{\sim} N(0, \sigma^2) \Rightarrow y_{ij} \sim N(\eta + \tau_i, \sigma^2) \rightarrow N(\mu_i, \sigma^2)$$

H_0 : There is no difference between treatments, $\tau_1 = \dots = \tau_k$

H_1 : There is difference between the treatments, i.e., not all τ_i are the same.

Use ANOVA decomposition

$$\begin{aligned} (y_{ij} - \bar{y}_{..})^2 &= \sum_i \sum_j (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_i \sum_j (y_{ij} - \bar{y}_{i.})^2 + 2 \sum_i \sum_j (\bar{y}_{i.} - \bar{y}_{..})(y_{ij} - \bar{y}_{i.}) \\ &= \sum_i \sum_j (\bar{y}_{i.} - \bar{y}_{..})^2 + \sum_i \sum_j (y_{ij} - \bar{y}_{i.})^2 + 2 \underbrace{\sum_i (\bar{y}_{i.} - \bar{y}_{..}) \sum_j^{n_i} (y_{ij} - \bar{y}_{i.})}_{=0} \end{aligned}$$

$$\frac{SS_i}{\sigma^2} \sim \chi_{k-1}^2$$

$$\hat{\sigma}^2 = \text{MSE}$$

$$\text{Rank}(X^\top X) = \text{Rank}(X)$$

Drop τ_1 from the parameters. then $\mathbf{X}$ exists. then $\eta = \mathbb{E}(y_{1j})$ and $\mathbb{E}(y_{2j}) - \mathbb{E}(y_{1j}) = \eta + \tau_2 - \eta = \tau_2$

$$\tau_i = \mathbb{E}(y_{ij}) - \mathbb{E}(y_{1j})$$

Estimates under the zero-sum constraint are

$$\hat{\eta} = \bar{y}_{..}, \hat{\tau}_i = \bar{y}_{i.} - \bar{y}_{..} \quad i = 1, \dots, k$$

$$\begin{aligned} \text{Var}(\bar{y}_{j.} - \bar{y}_{i.}) &= \text{Var}(\bar{y}_{i.}) + \text{Var}(\bar{y}_{j.}) - 2 \text{Cov}(\bar{y}_{i.}, \bar{y}_{j.}) \\ &= \frac{\sigma^2}{n_i} + \frac{\sigma^2}{n_j} - 0 \end{aligned}$$

$$df = N - k$$